

B.TECH UNIVERSITY EXAMINATION

Subject: Artificial Intelligence (AI) and Machine Learning

Duration: 3 Hours

Max Marks: 100

Instructions:

- Answer ALL questions in Section A.
- Answer any TWO questions from each of the remaining sections (B, C, D).
- All questions carry equal marks as indicated.
- Neat diagrams must be drawn wherever necessary.

SECTION A - Short Answer Questions

(4 marks each)

1. Define Artificial Intelligence and explain its applications.
2. Explain the difference between supervised and unsupervised learning.
3. What are production systems in AI?
4. Discuss the role of search strategies in AI problem-solving.
5. Explain the concept of reinforcement learning.

SECTION B - Long Answer Questions

(10 marks each)

1. Describe the architecture of an expert system with a neat diagram.
2. Explain decision trees and how they are used in machine learning.

SECTION C - Long Answer Questions

(10 marks each)

1. Discuss the various types of search algorithms used in AI with suitable examples.
2. Explain neural networks and their role in machine learning.

SECTION D - Long Answer Questions

(10 marks each)

1. Explain the concept of genetic algorithms and their applications in AI.
2. Discuss in detail the relationship between AI and Machine Learning with examples.